Paper 69 — Why Warmup the Learning Rate? Underlying Mechanisms and Improvements

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The paper in one minute

Question: Why does gradually increasing the learning rate help SGD and Adam, and can warmup be shortened?
Model/Method: Experiments across FCNs, ResNets and Transformers, with curvature/sharpness measurements and SGD/Adam variants.
Finding: The dominant empirical benefit is enabling larger *target* learning rates; early changes in ordinary or preconditioned sharpness make those rates tolerable. Catapult-informed starts and GI-Adam can reduce warmup needs.
Why we care: Warmup is partly a curvature-management intervention, not merely a ritual to "estimate early moments."

1. What is warmup really doing?

A typical schedule increases η_t from a small η_{init} to a desired η_{trgt} over T_{wrm} steps. Intuitively, the starting loss landscape may be too sharp for the target step size. Initial updates can move the model to a better-conditioned region.
The paper isolates this from another often-repeated claim, that Adam simply needs time to estimate enough moments. It finds that the ability to run a *larger final learning rate* explains most observed warmup benefits in its experiments. A longer warmup at a fixed target rate tends to have smaller additional value.

2. Worked example: curvature changes the allowable step

For $L(z) = \frac{1}{2}\lambda z^2$, the update factor is $1 - \eta\lambda$. The quadratic stability requirement is $\eta < 2/\lambda$. Suppose initially $\lambda = 20$, so $\eta < 0.1$ is the local threshold. A desired target rate $\eta = 0.3$ would be unstable in that fixed initial geometry. If early learning changes the local sharpness to $\lambda = 5$, the corresponding threshold becomes $2/5 = 0.4$, and 0.3 is locally admissible.
Interpretation: Warmup need not make the target step intrinsically smaller. It can help the network *change the landscape it encounters* before applying that target step. These numbers are illustrative, not experimental measurements from the paper.

3. Two warmup regimes and the catapult idea

The authors distinguish **progressive sharpening** (sharpness initially increases) from **natural sharpness reduction** (sharpness initially decreases). If learning rate rises enough to violate a local stability boundary, the loss may briefly spike and sharpness may subsequently fall; the authors call such behavior a *catapult*. Warmup can make this transition more gradual, and a loss-based line search can choose a more useful starting learning rate.
For adaptive Adam, let P_t denote the positive diagonal preconditioner implicit in the update $\Delta\theta = -\eta P_t g_t$. The curvature relevant to this update is the largest eigenvalue of

$$P_t^{1/2} H_t P_t^{1/2},$$

not simply $\lambda_{\max}(H_t)$. This *preconditioned sharpness* can start extremely high, even when the ordinary Hessian is not particularly sharp.

GI-Adam: a concrete modification

Initialize Adam's second moment with the first squared gradient, $v_0 = g_0^2$, rather than zero. The authors show this reduces initial preconditioned sharpness and can act like an implicit warmup under their bias-correction convention.

4. What experiments actually establish

Figures 1–4 compare warmup durations and achievable target learning rates across parameterizations, optimizers, models and datasets including CIFAR, TinyImageNet and WikiText. The main reported benefit is robust tolerance to larger η_{trgt}. For fixed target rate, improvements from longer warmup are usually smaller. In some configurations, a catapult-informed start shortens or eliminates warmup; GI-Adam improves failure boundaries in their tests.
These are *empirical mechanisms and algorithm proposals*, not a theorem guaranteeing all warmup schedules will improve training.

5. Why Adam needs a different curvature scale

Suppose $H = \text{diag}(20, 1)$ and an illustrative positive preconditioner is $P = \text{diag}(0.01, 1)$. Raw largest curvature is 20, but the transformed matrix is

$$P^{1/2} H P^{1/2} = \text{diag}(0.2, 1),$$

whose largest eigenvalue is 1. The curvature relevant to a preconditioned update must be assessed in *rescaled coordinates*. This explains why judging Adam by ordinary Hessian sharpness alone can mislead. The paper finds the reverse initial problem can also occur: Adam's preconditioned sharpness may start very high, motivating warmup or modified initialization.
The toy matrices are pedagogical, not experimental measurements.

5. Assumptions and limitations

Setting: Specific SGD and Adam implementations, architectures, parameterizations, initialization scales, and experimental stability criteria. Linear/quadratic thresholds are only local intuitions for nonlinear training.
Limits: There is no universal optimal warmup duration or general convergence theorem for arbitrary neural networks. Catapult effects and GI-Adam improvements depend on task/hyperparameters; extra warmup can still help for reasons not isolated in the selected experiments.

6. Connections to the survey

Li–Wang–Li (2022) supplies a progressive-sharpening explanation; **Kalra et al. (2025)** later studies sharpness regimes via fixed-point dynamics. **Galli (2026)** explores adaptive large steps and flat saddle regions, while this paper focuses on the *warmup path* to a larger target rate.

7. Quick reference

Concept	Meaning / equation
Linear warmup	$\eta_t : \eta_{\text{init}} \to \eta_{\text{trgt}}$
GD quadratic threshold	$\eta\lambda_{\max}(H) < 2$
Adam curvature	$\lambda_{\max}(P^{1/2}HP^{1/2})$
GI-Adam initialization	$v_0 = g_0^2$
Original to read	Sections 3–5, Figures 1–4

8. Papers Master note

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Finds empirically that warmup chiefly permits larger target learning rates by managing ordinary or preconditioned sharpness. Distinguishes progressive sharpening from initial sharpness reduction; proposes catapult-informed starts and GI-Adam. No universal training guarantee.

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